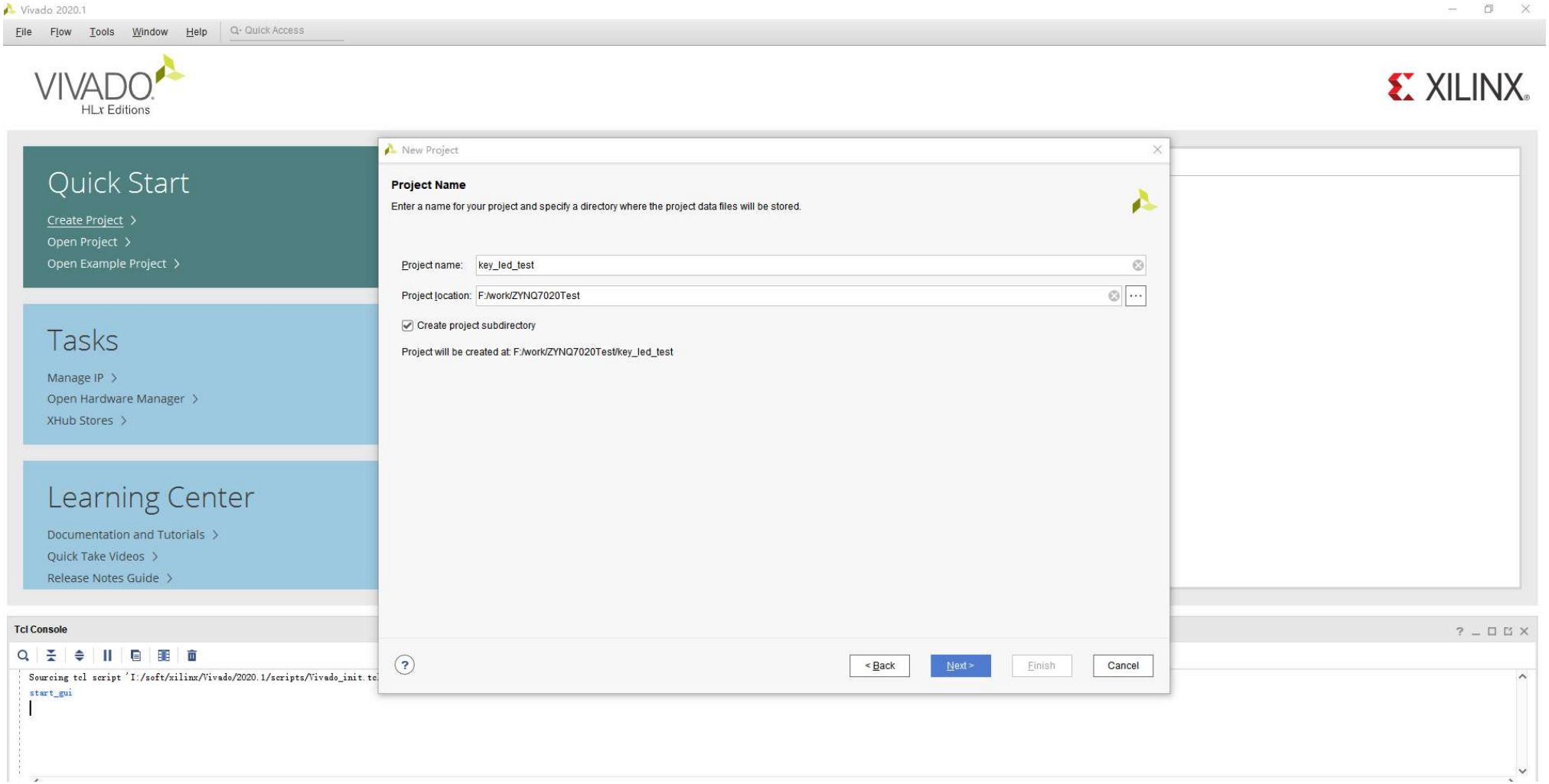
ZYNQ7010 开发板烧录教程

环境：

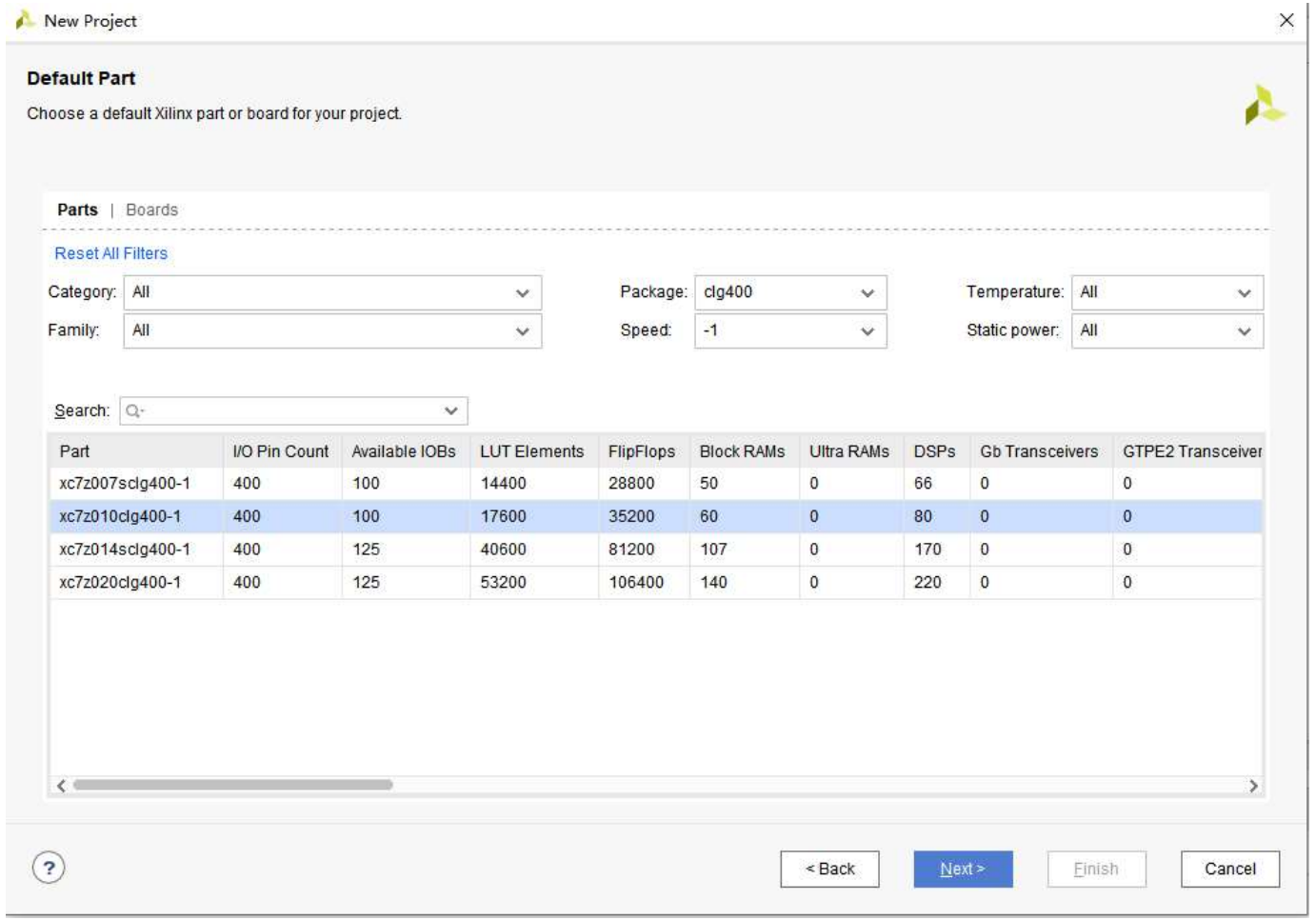
Vivado 2020.1

创建工程

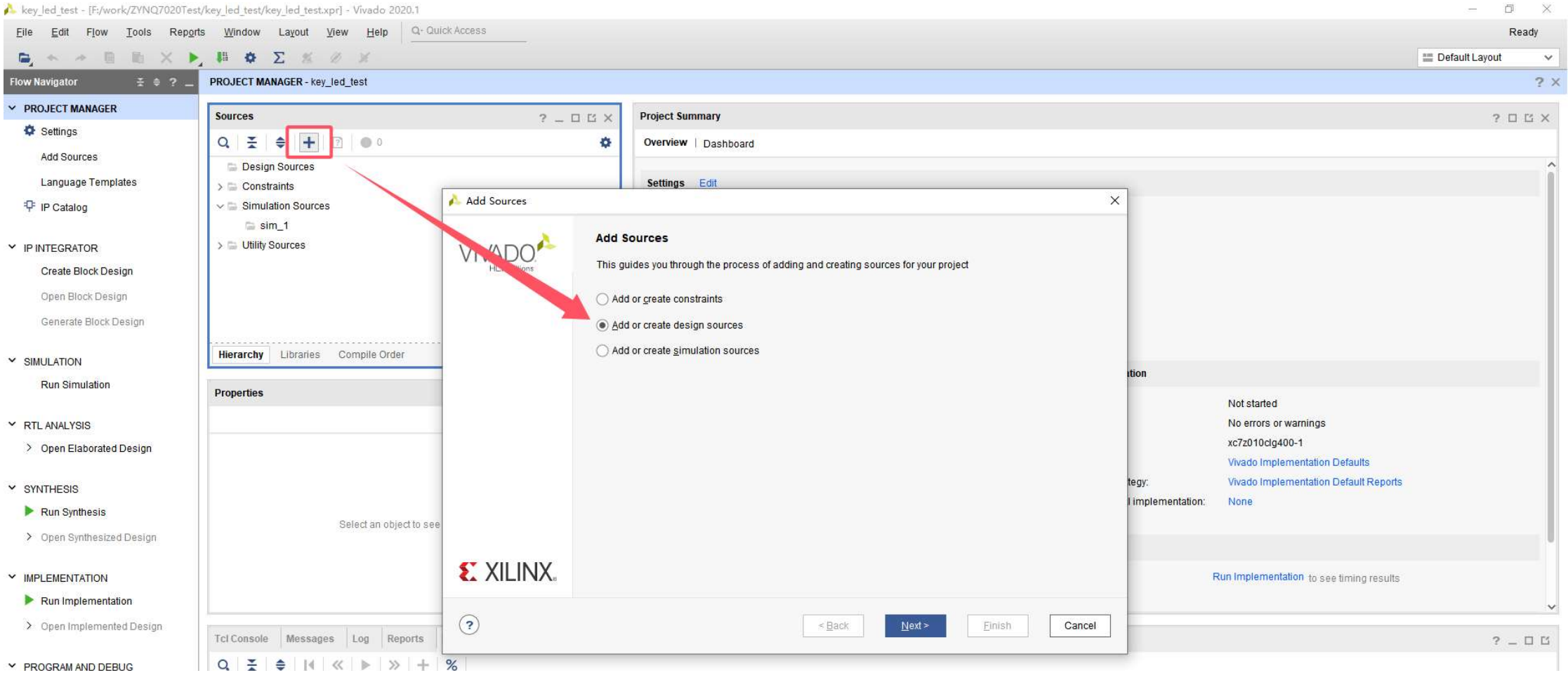

Vivado 2020.1
File Flow Tools Window Help Quick Access
VIVADO HLx Editions
XILINX
Quick Start
Create Project
Open Project
Open Example Project
Tasks
Manage IP
Open Hardware Manager
XHub Stores
Learning Center
Documentation and Tutorials
Quick Take Videos
Release Notes Guide
New Project
Project Name
Enter a name for your project and specify a directory where the project data files will be stored.
Project name: key_led_test
Project location: F:/work/ZYNQ7020Test
Create project subdirectory
Project will be created at: F:/work/ZYNQ7020Test/key_led_test
< Back
Next >
Finish
Cancel
Tcl Console
Sourcing tcl script 'I:/soft/xilinx/Vivado/2020.1/scripts/Vivado_init.tcl'
start_gui

选择芯片

- 封装选择clg400
- 速度等级-1

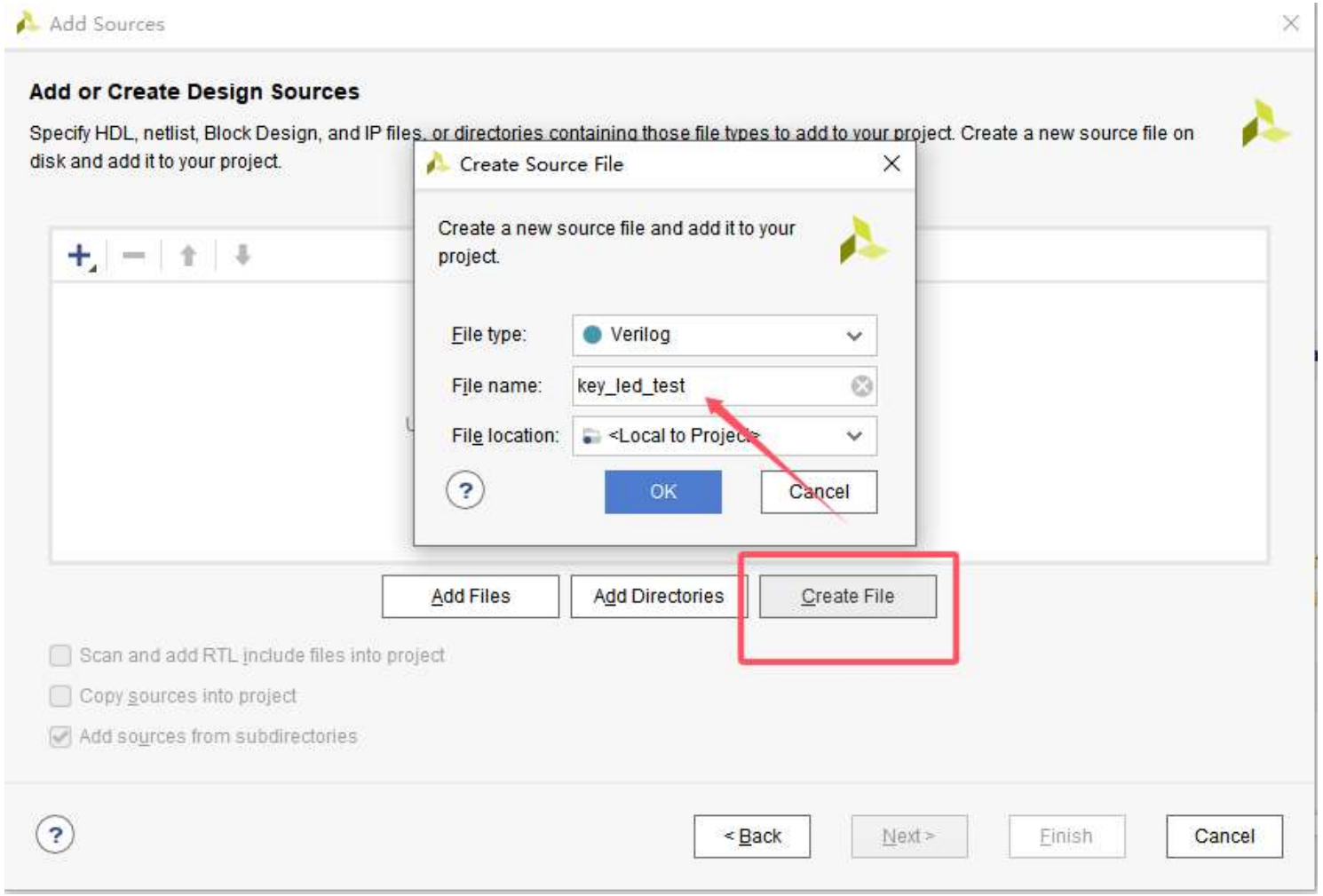


创建verilog源文件



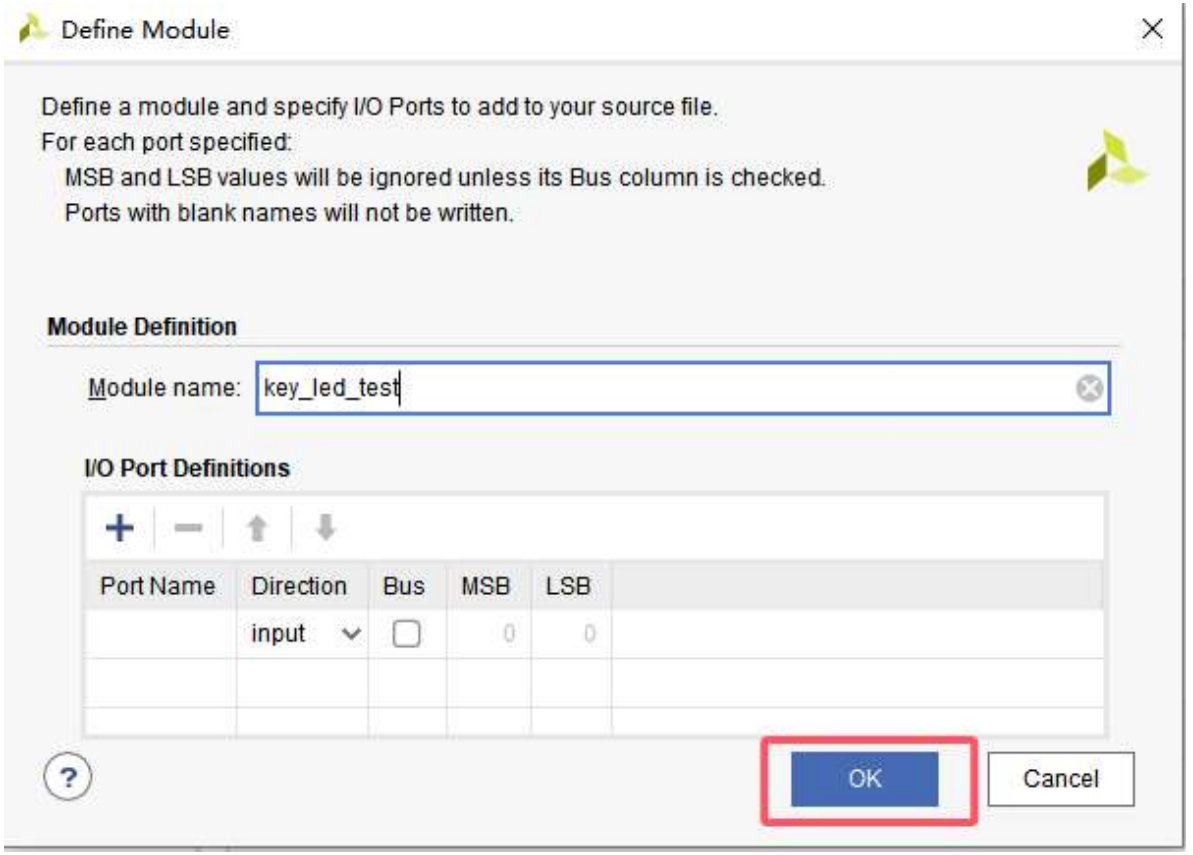
输入文件名

- 默认路径为<prj>\<prj>.srcs\sources_1\new



确认输入输出端口

- 此处不需要指定端口，之后可以在源文件内指定
- 与源文件开头的input与output效果相同



简单编写亮灯程序

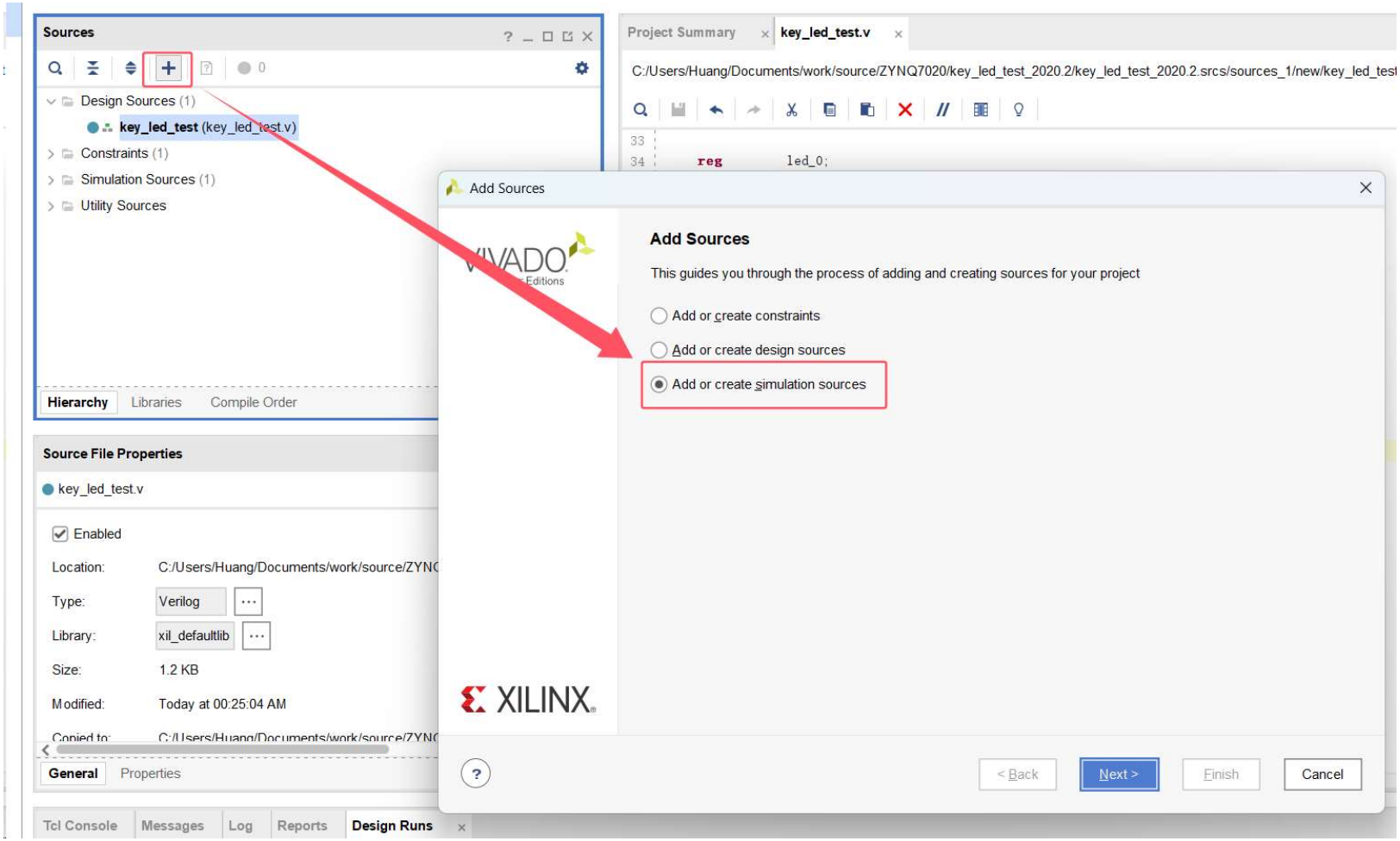
- 功能：led0默认灭，led1默认亮，按下key0时led0亮，按下key1时led1灭
- 注意按键为低电平有效，未按下时为高电平

key_led_test.v

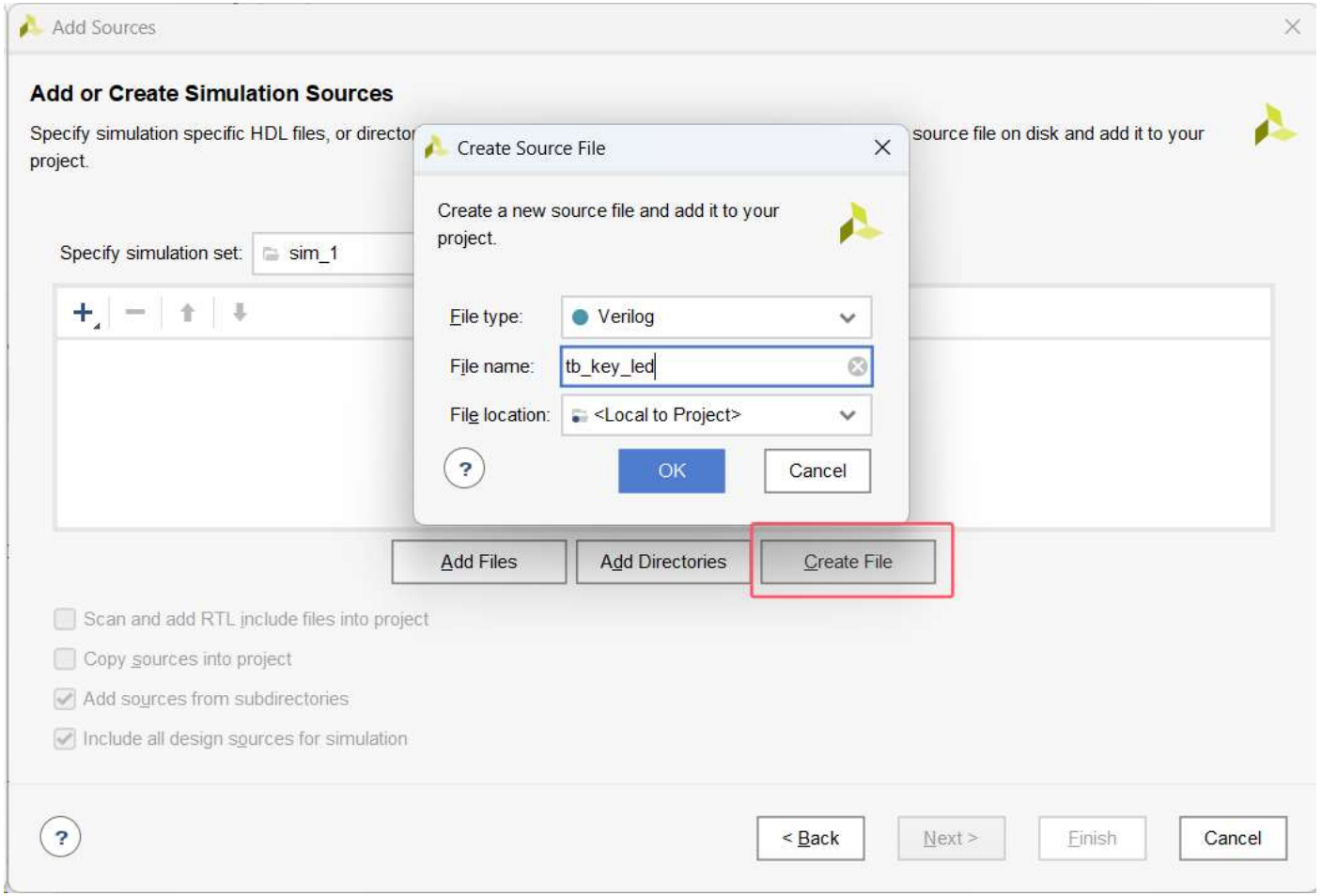
F:/work/ZYNQ7020Test/key_led_test/key_led_test.srcs/sources_1/new/key_led_test.v

```
20  //////////////////////////////////////////////////////////////////////////////////
21
22
23  module key_led_test(
24      input       clk_i,
25      input       rst_n,
26
27      input       key_0_i,
28      input       key_1_i,
29
30      output      led_0_o,
31      output      led_1_o
32      );
33
34      reg         led_0;
35      reg         led_1;
36
37      always @(posedge clk_i or negedge rst_n) begin
38          if(~rst_n)
39              led_0 <= 1'b0;
40          else
41              if(~key_0_i)
42                  led_0   <= 1'b1;
43              else
44                  led_0   <= 1'b0;
45      end
46
47
48
49      always @(posedge clk_i or negedge rst_n) begin
50          if(~rst_n)
51              led_1 <= 1'b1;
52          else
53              if(~key_1_i)
54                  led_1   <= 1'b0;
55              else
56                  led_1   <= 1'b1;
57      end
58
59      assign led_0_o = led_0;
60      assign led_1_o = led_1;
61
62  endmodule
63
```

创建仿真文件


Sources
Design Sources (1)
key_led_test (key_led_test.v)
Constraints (1)
Simulation Sources (1)
Utility Sources
Hierarchy
Libraries
Compile Order
Source File Properties
key_led_test.v
Enabled
Location:
Type:
Verilog
Library:
xil_defaultlib
Size:
1.2 KB
Modified:
Today at 00:25:04 AM
General
Properties
Tcl Console
Messages
Log
Reports
Design Runs
Project Summary
key_led_test.v
C:/Users/Huang/Documents/work/source/ZYNQ7020/key_led_test_2020.2/key_led_test_2020.2.srcs/sources_1/new/key_led_test
reg led_0;
Add Sources
Add Sources
This guides you through the process of adding and creating sources for your project
Add or create constraints
Add or create design sources
Add or create simulation sources
XILINX
< Back
Next >
Finish
Cancel

指定文件名


Add Sources
Add or Create Simulation Sources
Specify simulation specific HDL files, or director
source file on disk and add it to your project.
Specify simulation set:
sim_1
Create Source File
Create a new source file and add it to your project.
File type:
Verilog
File name:
tb_key_led
File location:
<Local to Project>
OK
Cancel
Add Files
Add Directories
Create File
Scan and add RTL include files into project
Copy sources into project
Add sources from subdirectories
Include all design sources for simulation
< Back
Next >
Finish
Cancel

简单编写仿真激励

tb_key_led.v

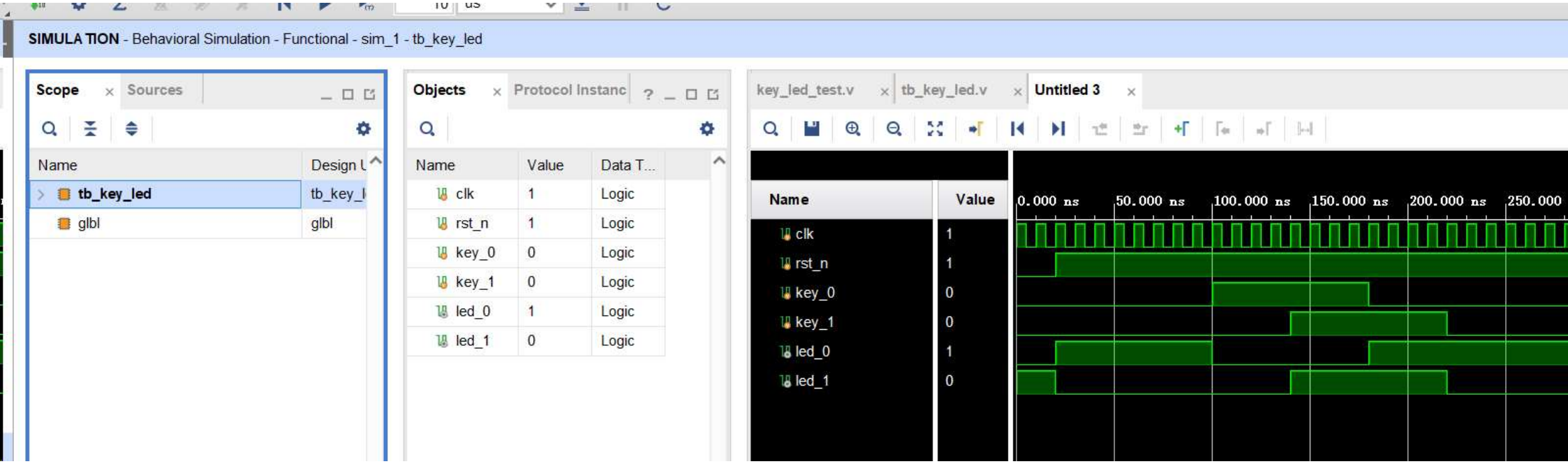
C:/Users/Huang/Documents/work/source/ZYNQ7020/key_led_test_2020.2/key_led_test_2020.2.srcs/sim_1/ne

```
22
23  module tb_key_led();
24
25      reg     clk;
26      reg     rst_n;
27
28      reg     key_0;
29      reg     key_1;
30
31      wire    led_0;
32      wire    led_1;
33
34      always #5 clk = ~clk;
35
36      initial begin
37          clk = 1'b1;
38          rst_n = 1'b0;
39
40          key_0 = 1'b0;
41          key_1 = 1'b0;
42
43          #20
44          rst_n = 1'b1;
45
46          #80
47          key_0 = 1'b1;
48
49          #40
50          key_1 = 1'b1;
51
52          #40
53          key_0 = 1'b0;
54
55          #40
56          key_1 = 1'b0;
57      end
58
59      key_led_test u_key_led_test(
60          .clk_i(clk),
61          .rst_n(rst_n),
62
63          .key_0_i(key_0),
64          .key_1_i(key_1),
65
66          .led_0_o(led_0),
67          .led_1_o(led_1)
68      );
69
70  endmodule
71
```

启动仿真

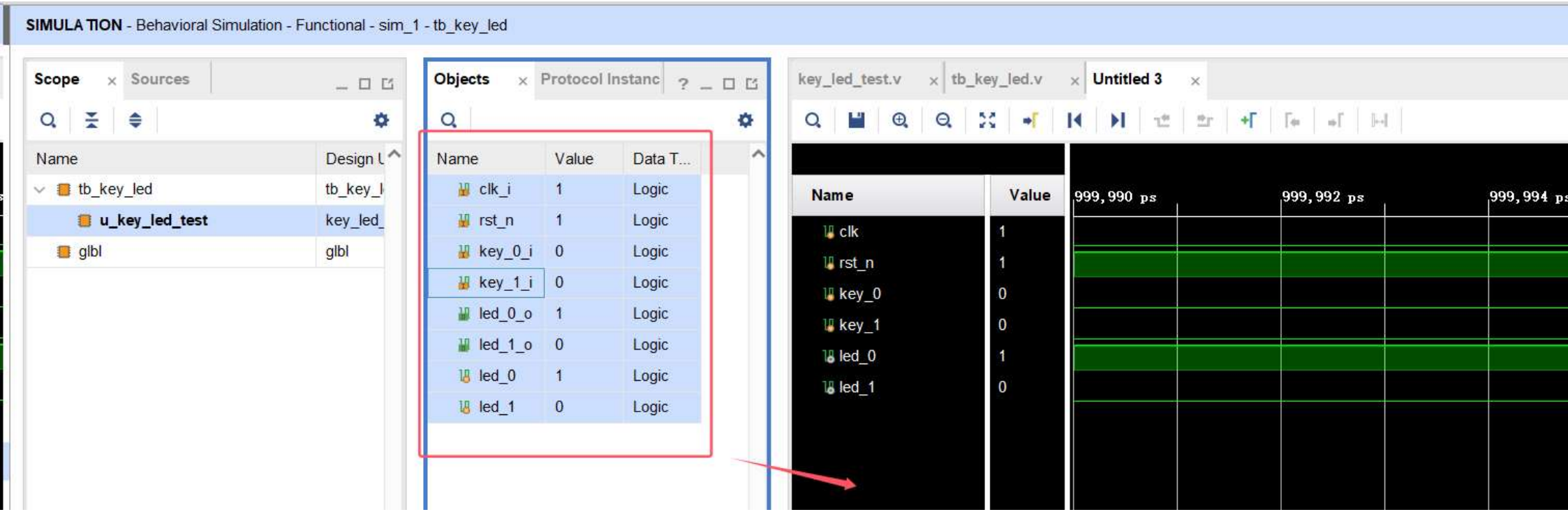

SIMULATION
Run Simulation
Hierarchy
Libraries
Run Behavioral Simulation
RTL ANALY
Run Post-Synthesis Functional Simulation
Open El
Run Post-Synthesis Timing Simulation
Run Post-Implementation Functional Simulation
SYNTHESIS
Run Post-Implementation Timing Simulation
Run Synthesis

查看波形是否符合需求


SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_key_led
Scope
Sources
Name
Design U
tb_key_led
tb_key_l
glbl
glbl
Objects
Protocol Instanc
Name
Value
Data T...
clk
1
Logic
rst_n
1
Logic
key_0
0
Logic
key_1
0
Logic
led_0
1
Logic
led_1
0
Logic
key_led_test.v
tb_key_led.v
Untitled 3
Name
Value
clk
1
rst_n
1
key_0
0
key_1
0
led_0
1
led_1
0
0.000 ns
50.000 ns
100.000 ns
150.000 ns
200.000 ns
250.000

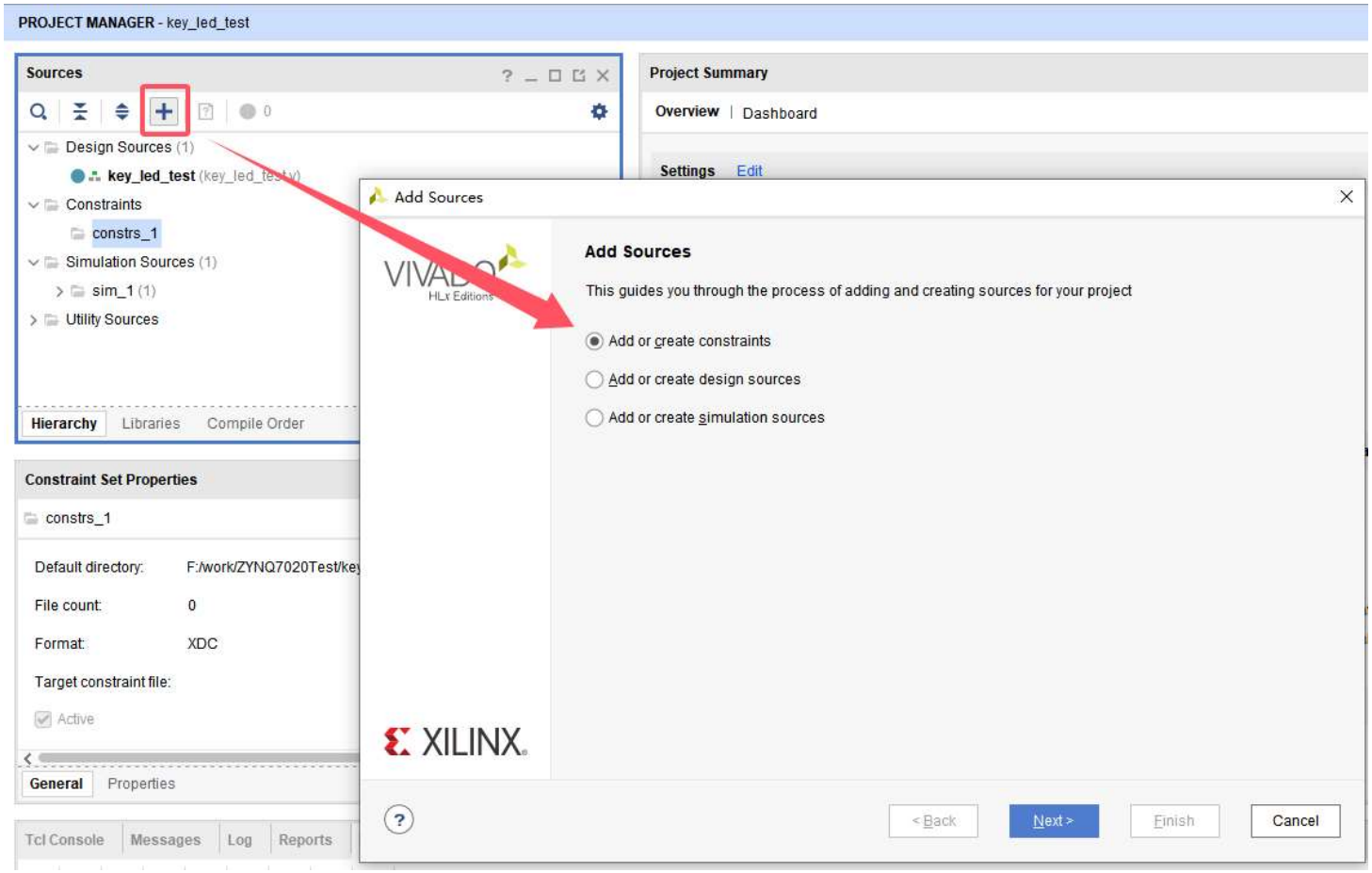
查看模块内的信号

- 点开模块，将需要观察的信号拖到波形图中，重启仿真

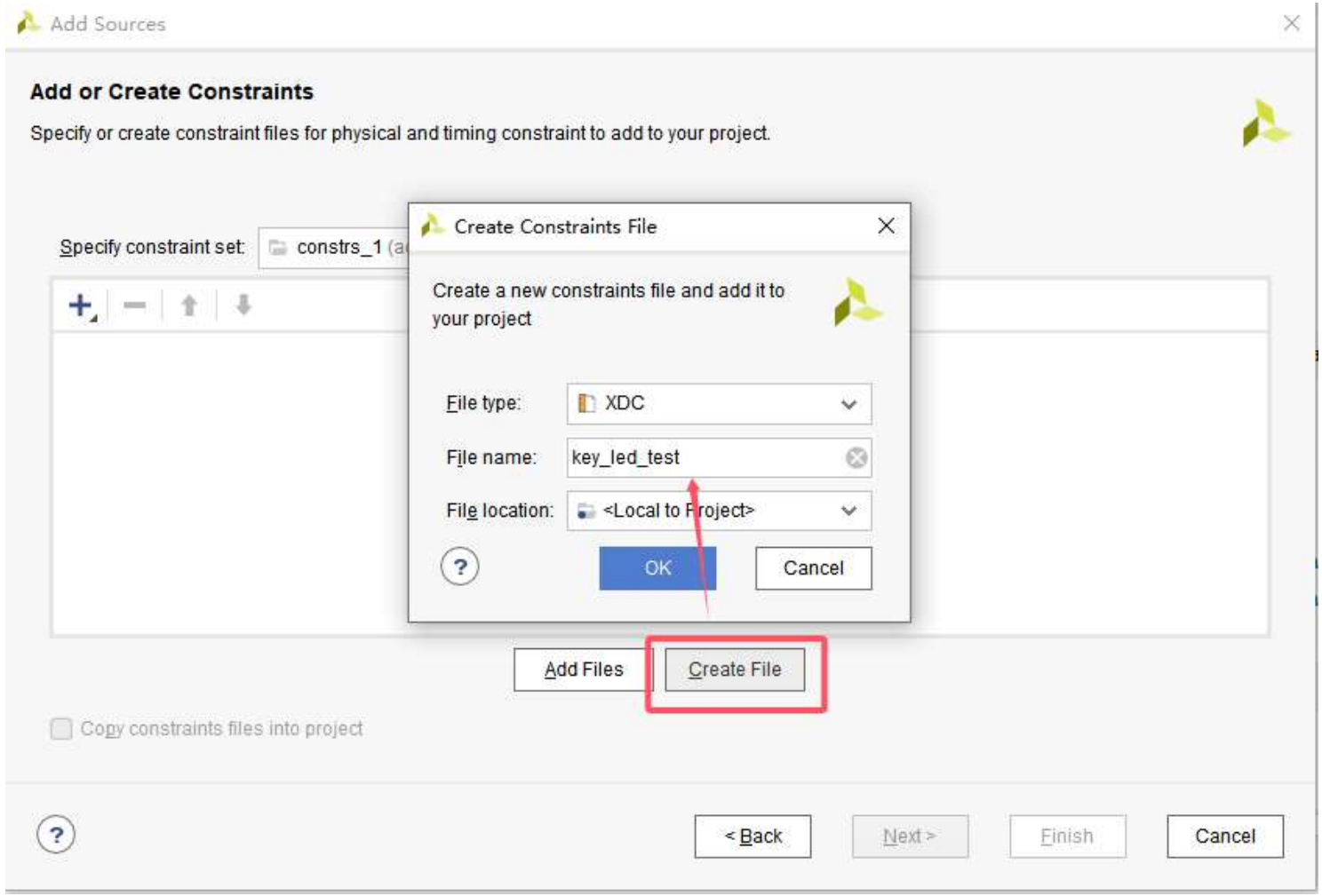


创建约束文件

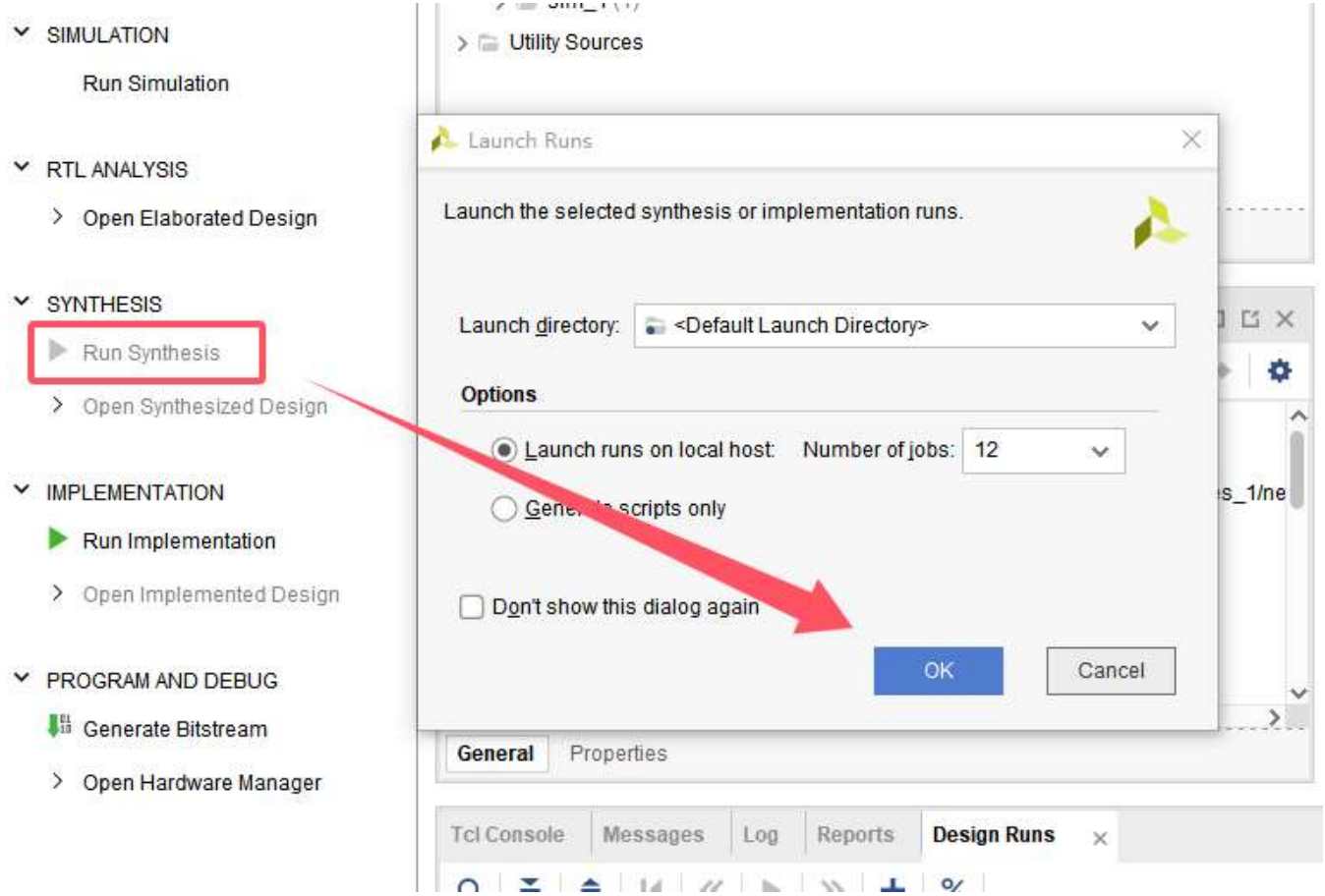
- 约束文件指定输入输出端口与实际IO口的关系
- 约束文件指定时钟频率，用于计算时序



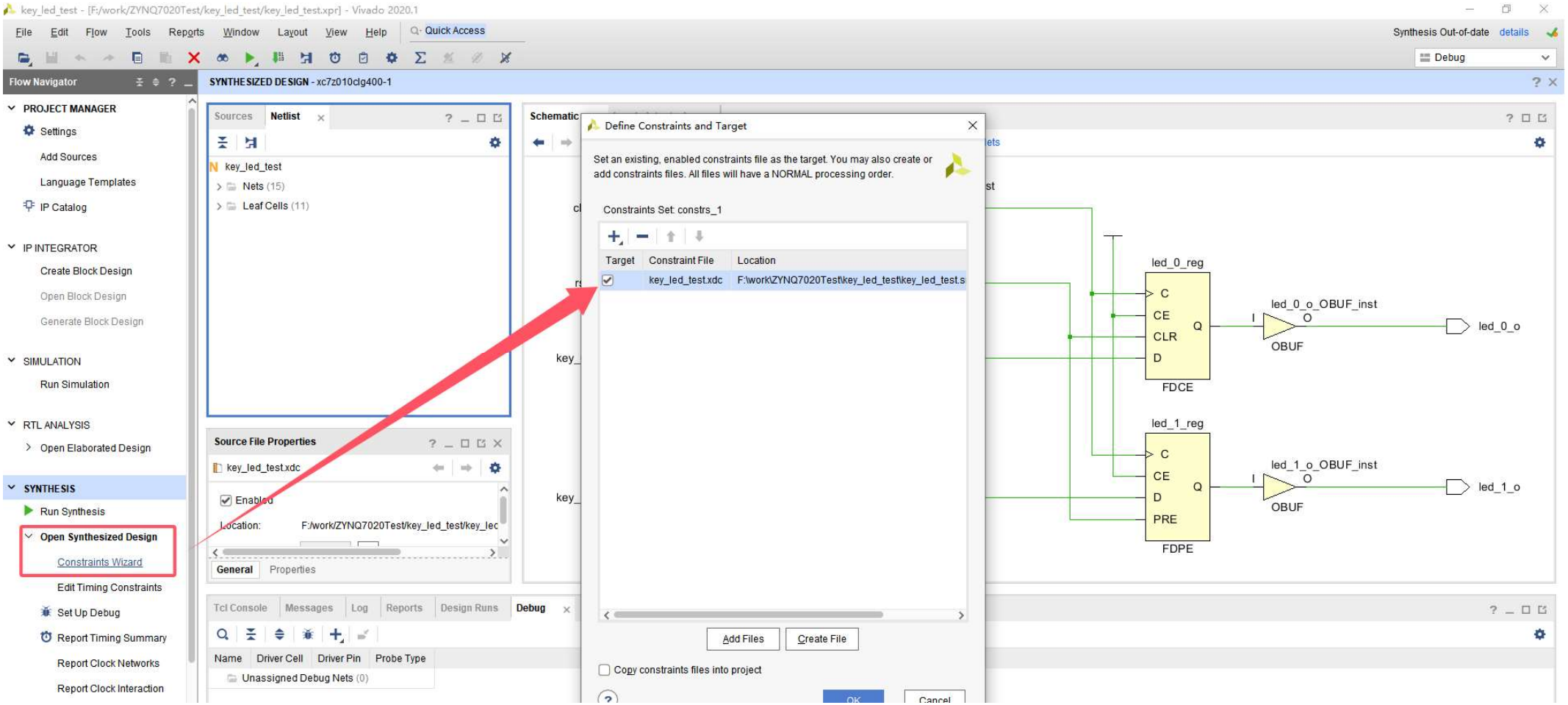
输入文件名


Add Sources
Add or Create Constraints
Specify or create constraint files for physical and timing constraint to add to your project.
Specify constraint set:
constrs_1
Create Constraints File
Create a new constraints file and add it to your project
File type:
XDC
File name:
key_led_test
File location:
<Local to Project>
OK
Cancel
Add Files
Create File
Copy constraints files into project
< Back
Next >
Finish
Cancel

启动综合

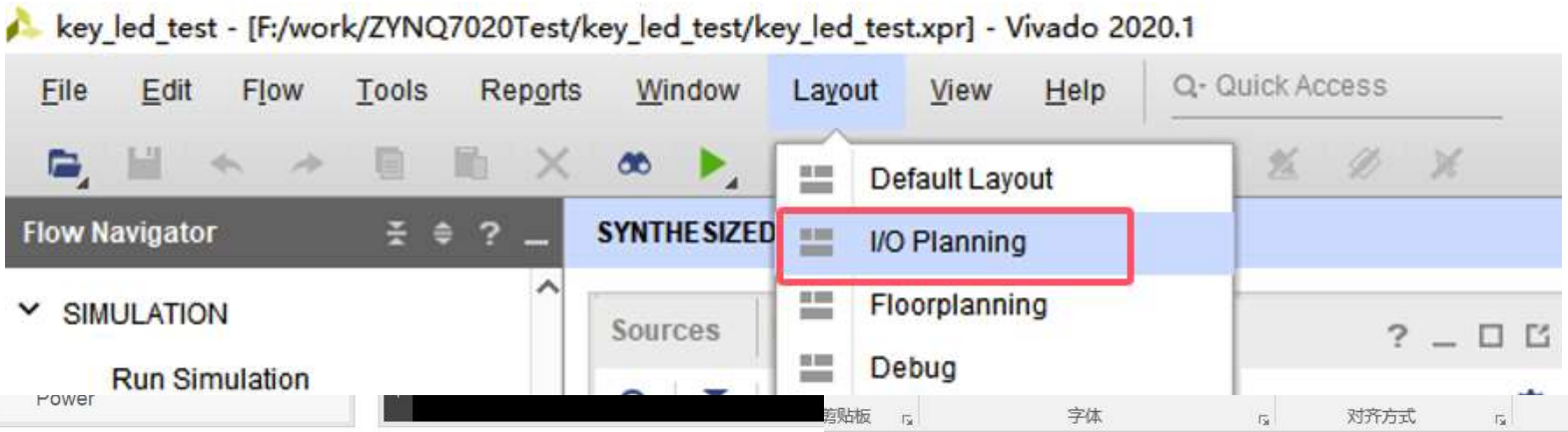

SIMULATION
Run Simulation
RTL ANALYSIS
Open Elaborated Design
SYNTHESIS
Run Synthesis
Open Synthesized Design
IMPLEMENTATION
Run Implementation
Open Implemented Design
PROGRAM AND DEBUG
Generate Bitstream
Open Hardware Manager
Utility Sources
Launch Runs
Launch the selected synthesis or implementation runs.
Launch directory:
<Default Launch Directory>
Options
Launch runs on local host:
Number of jobs:
12
Don't show this dialog again
OK
Cancel
General
Properties
Tcl Console
Messages
Log
Reports
Design Runs

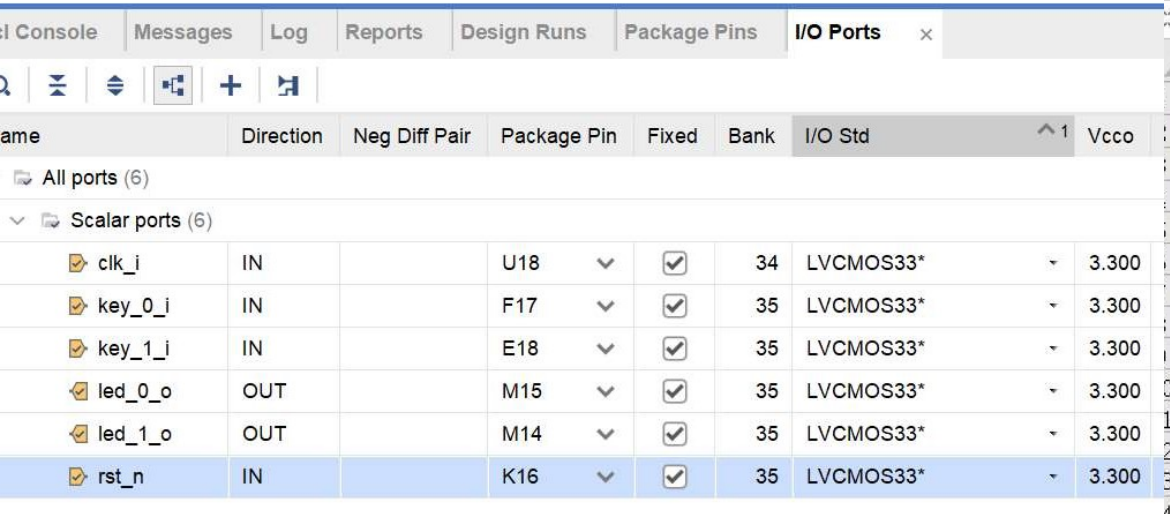
指定已创建的xdc文件



指定IO管脚与电平

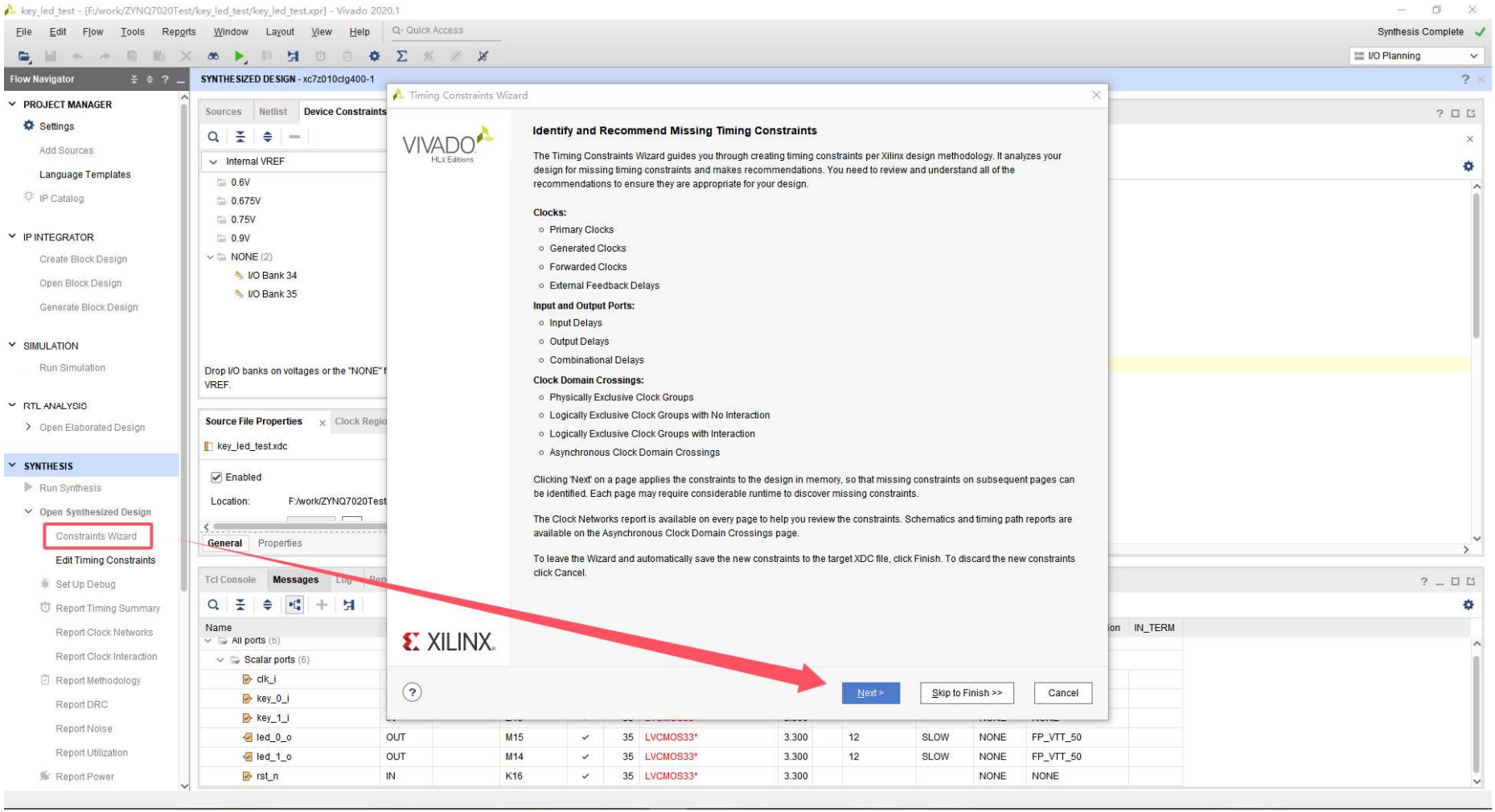
- 找不到的，双击Open Synthesized Design使这行字变黑后再点开Layout



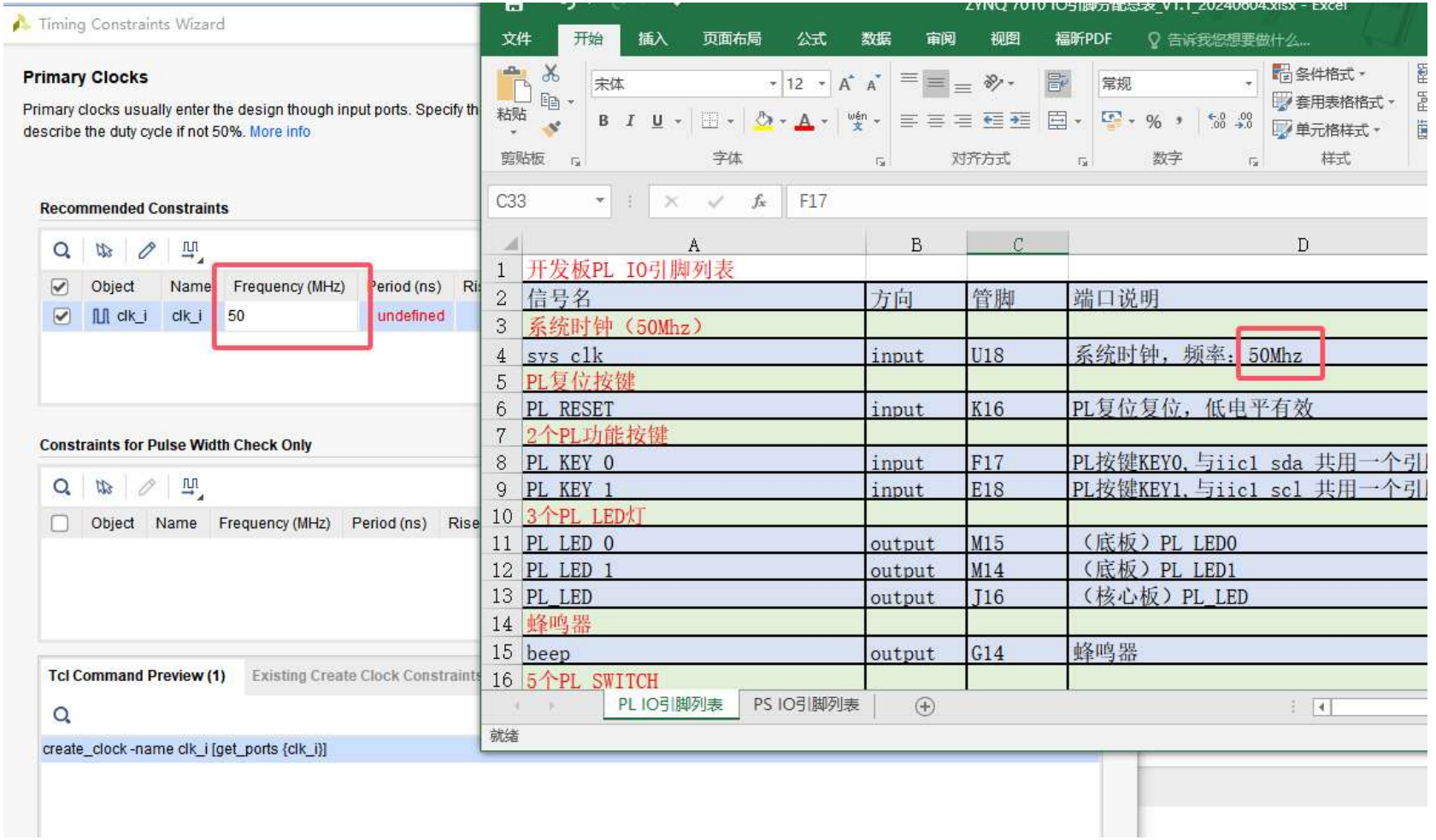




添加时序约束

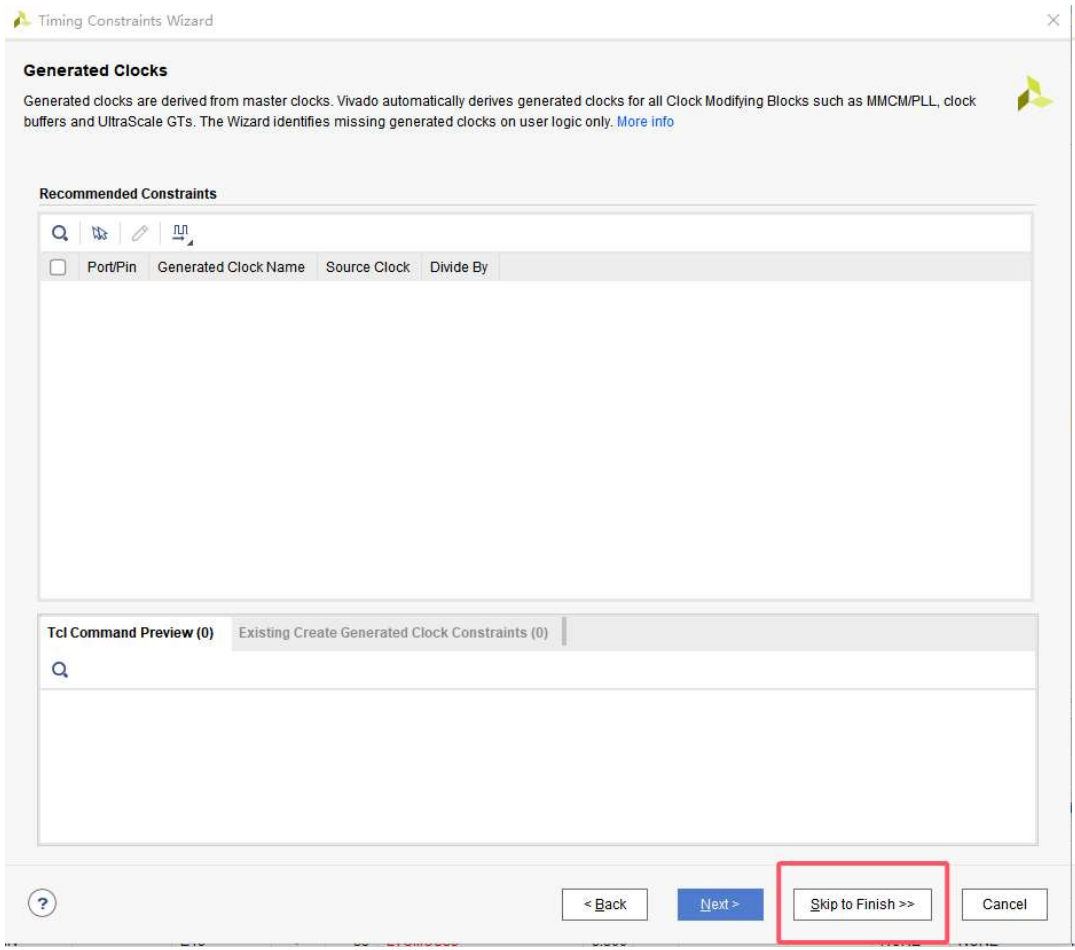

key_led_test - [F:/work/ZYNQ7020Test/key_led_test/key_led_test.xpr] - Vivado 2020.1
File Edit Flow Tools Reports Window Layout View Help
Quick Access
Synthesis Complete
I/O Planning
Flow Navigator
SYNTHESIZED DESIGN - xc7z010clg400-1
PROJECT MANAGER
Settings
Add Sources
Language Templates
IP Catalog
IP INTEGRATOR
Create Block Design
Open Block Design
Generate Block Design
SIMULATION
Run Simulation
RTL ANALYSIS
Open Elaborated Design
SYNTHESIS
Run Synthesis
Open Synthesized Design
Constraints Wizard
Edit Timing Constraints
Set Up Debug
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Noise
Report Utilization
Report Power
Sources
Netlist
Device Constraints
Internal VREF
0.6V
0.675V
0.75V
0.9V
NONE (2)
I/O Bank 34
I/O Bank 35
Drop I/O banks on voltages or the "NONE" f
VREF.
Source File Properties
key_led_test.xdc
Enabled
Location:
F:/work/ZYNQ7020Test
General
Properties
Tcl Console
Messages
Name
All ports (6)
Scalar ports (6)
clk_i
key_0_i
key_1_i
led_0_o
led_1_o
rst_n
IN_TERM
Timing Constraints Wizard
VIVADO
HLx Editions
Identify and Recommend Missing Timing Constraints
The Timing Constraints Wizard guides you through creating timing constraints per Xilinx design methodology. It analyzes your design for missing timing constraints and makes recommendations. You need to review and understand all of the recommendations to ensure they are appropriate for your design.
Clocks:
Primary Clocks
Generated Clocks
Forwarded Clocks
External Feedback Delays
Input and Output Ports:
Input Delays
Output Delays
Combinational Delays
Clock Domain Crossings:
Physically Exclusive Clock Groups
Logically Exclusive Clock Groups with No Interaction
Logically Exclusive Clock Groups with Interaction
Asynchronous Clock Domain Crossings
Clicking 'Next' on a page applies the constraints to the design in memory, so that missing constraints on subsequent pages can be identified. Each page may require considerable runtime to discover missing constraints.
The Clock Networks report is available on every page to help you review the constraints. Schematics and timing path reports are available on the Asynchronous Clock Domain Crossings page.
To leave the Wizard and automatically save the new constraints to the target XDC file, click Finish. To discard the new constraints click Cancel.
XILINX
Next >
Skip to Finish >>
Cancel

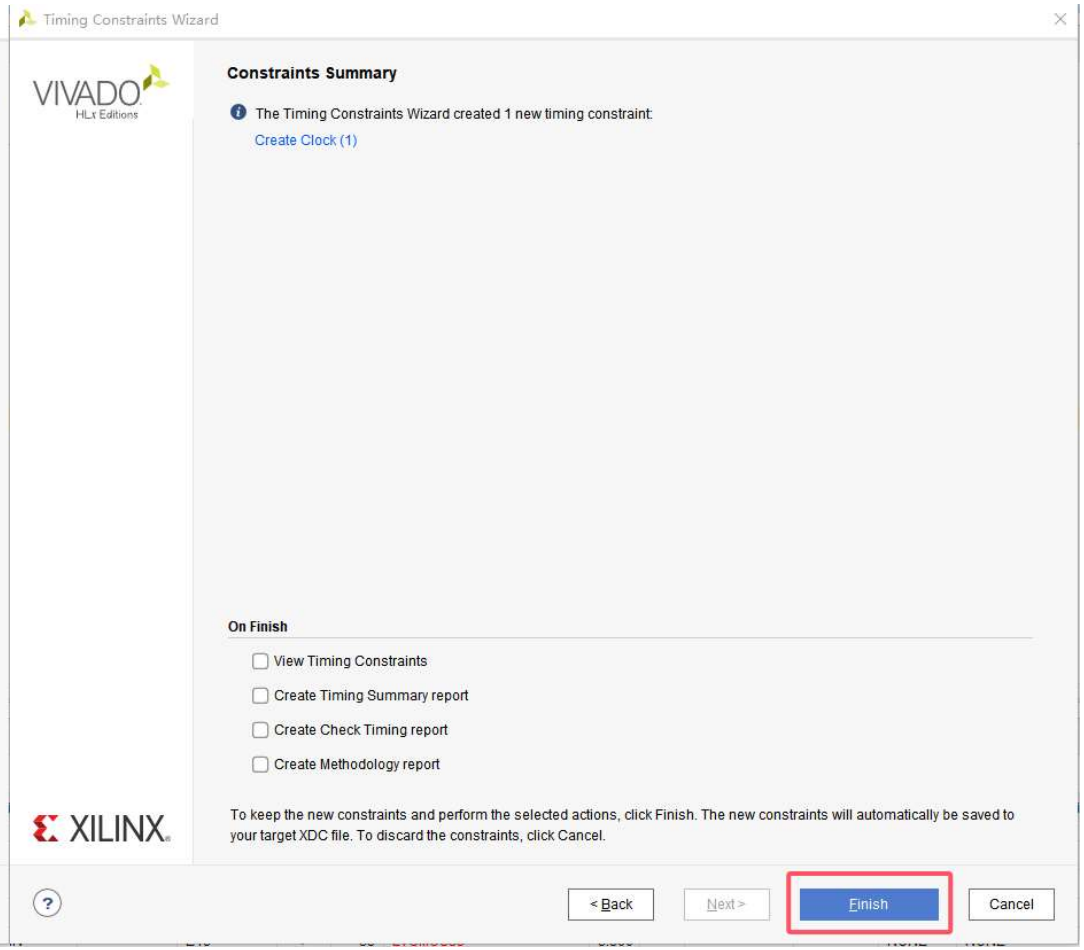
指定主时钟频率



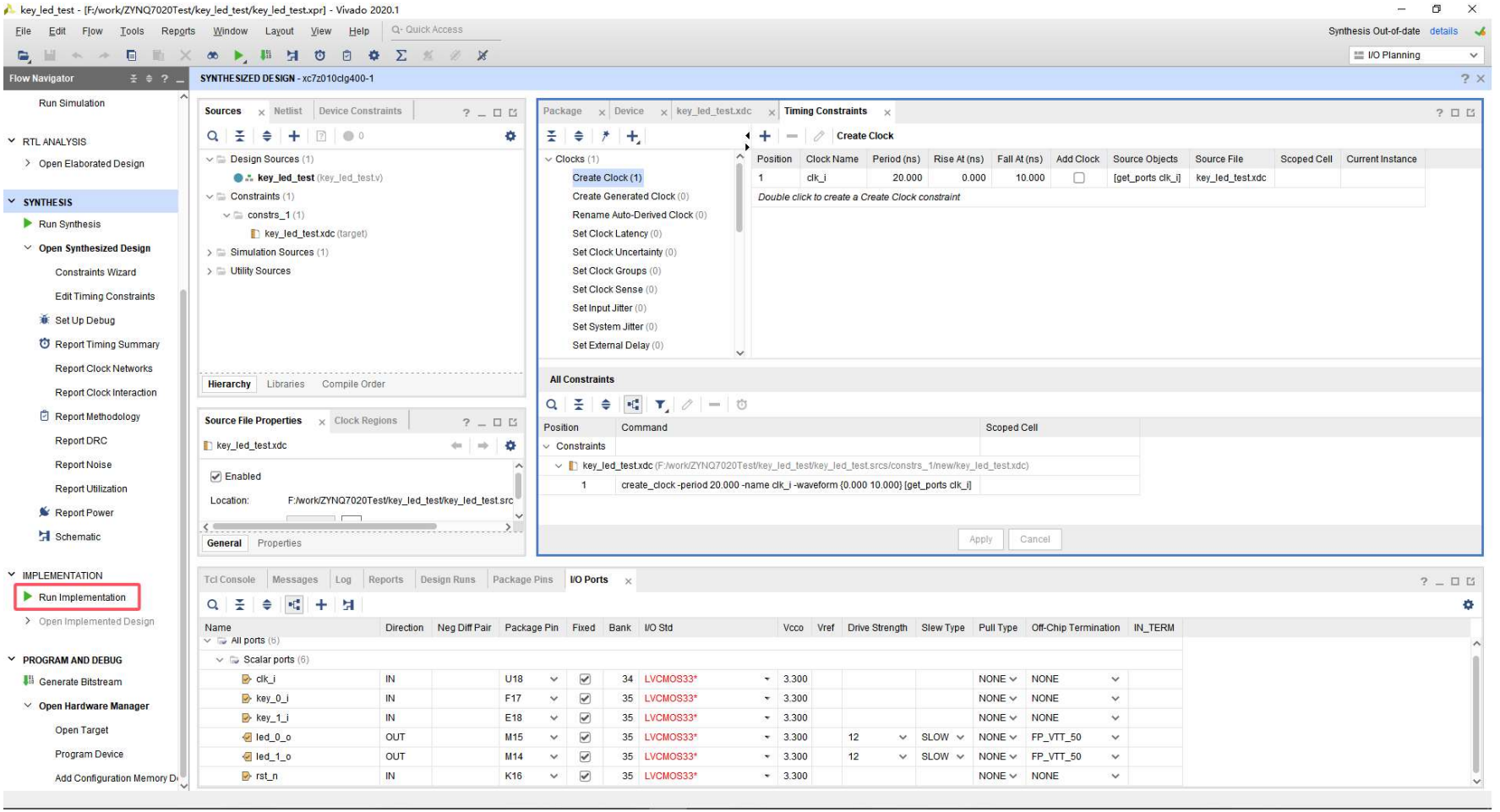
	A	B	C	D
1	开发板PL IO引脚列表			
2	信号名	方向	管脚	端口说明
3	系统时钟（50Mhz）			
4	sys_clk	input	U18	系统时钟，频率：50Mhz
5	PL复位按键			
6	PL_RESET	input	K16	PL复位复位，低电平有效
7	2个PL功能按键			
8	PL_KEY_0	input	F17	PL按键KEY0,与iic1_sda 共用一个引
9	PL_KEY_1	input	E18	PL按键KEY1,与iic1_scl 共用一个引
10	3个PL_LED灯			
11	PL_LED_0	output	M15	（底板）PL_LED0
12	PL_LED_1	output	M14	（底板）PL_LED1
13	PL_LED	output	J16	（核心板）PL_LED
14	蜂鸣器			
15	beep	output	G14	蜂鸣器
16	5个PL_SWITCH			

跳过剩下所有步骤

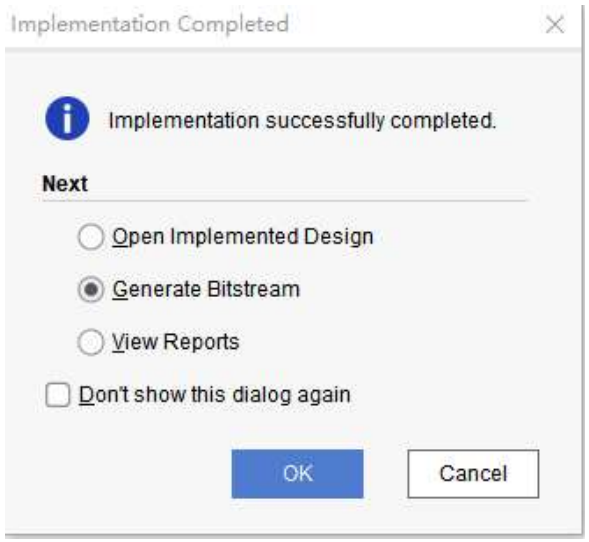

Timing Constraints Wizard
Generated Clocks
Generated clocks are derived from master clocks. Vivado automatically derives generated clocks for all Clock Modifying Blocks such as MMCM/PLL, clock buffers and UltraScale GTs. The Wizard identifies missing generated clocks on user logic only. More info
Recommended Constraints
Port/Pin
Generated Clock Name
Source Clock
Divide By
Tcl Command Preview (0)
Existing Create Generated Clock Constraints (0)
< Back
Next >
Skip to Finish >>
Cancel


Timing Constraints Wizard
VIVADO
HLx Editions
Constraints Summary
The Timing Constraints Wizard created 1 new timing constraint:
Create Clock (1)
On Finish
View Timing Constraints
Create Timing Summary report
Create Check Timing report
Create Methodology report
XILINX
To keep the new constraints and perform the selected actions, click Finish. The new constraints will automatically be saved to your target XDC file. To discard the constraints, click Cancel.
< Back
Next >
Finish
Cancel

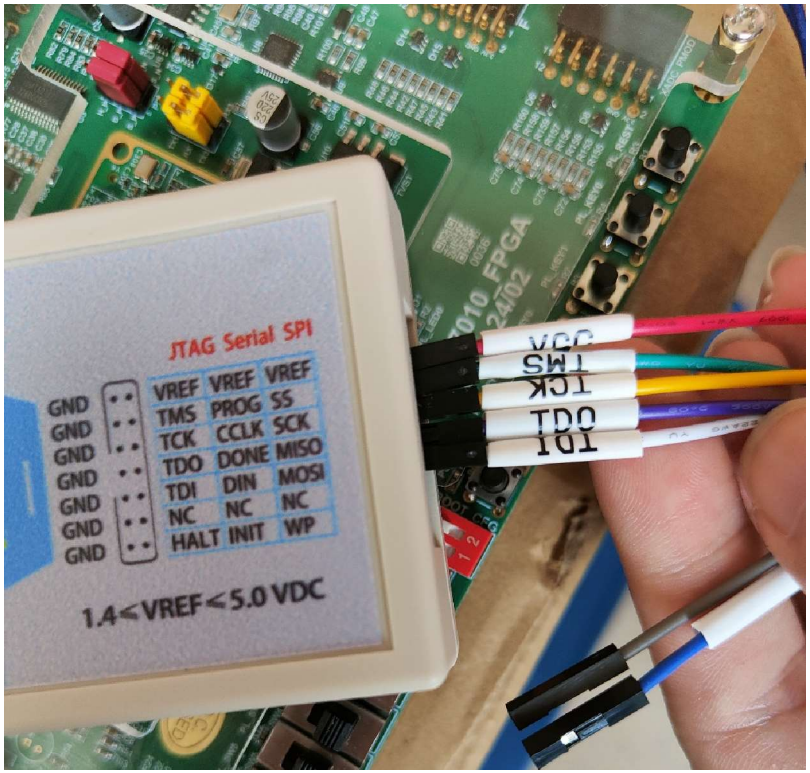
检查一遍，Ctrl-S后启动布线



布线没有报错可继续生成比特流



连接板卡与仿真器

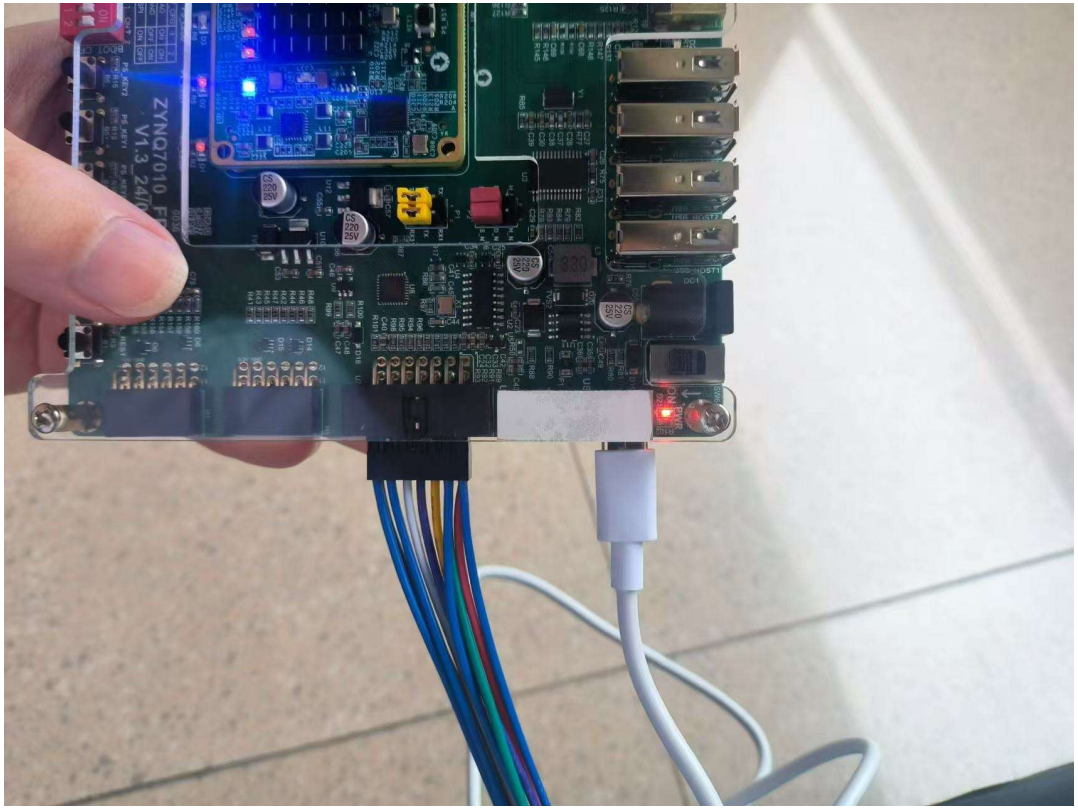


连接下载器时注意线序，根据线套上的指示与下载器上的接口指示对应连接。

注意：

- 按照最左侧的线序指引进行连线
- 杜邦线上的VCC对应下载器上的VREF
- 图上连接的线位于接口靠下一排(以贴纸为上方向)
- 地线也需要连接，最终所有的杜邦线都应该连接到下载器上

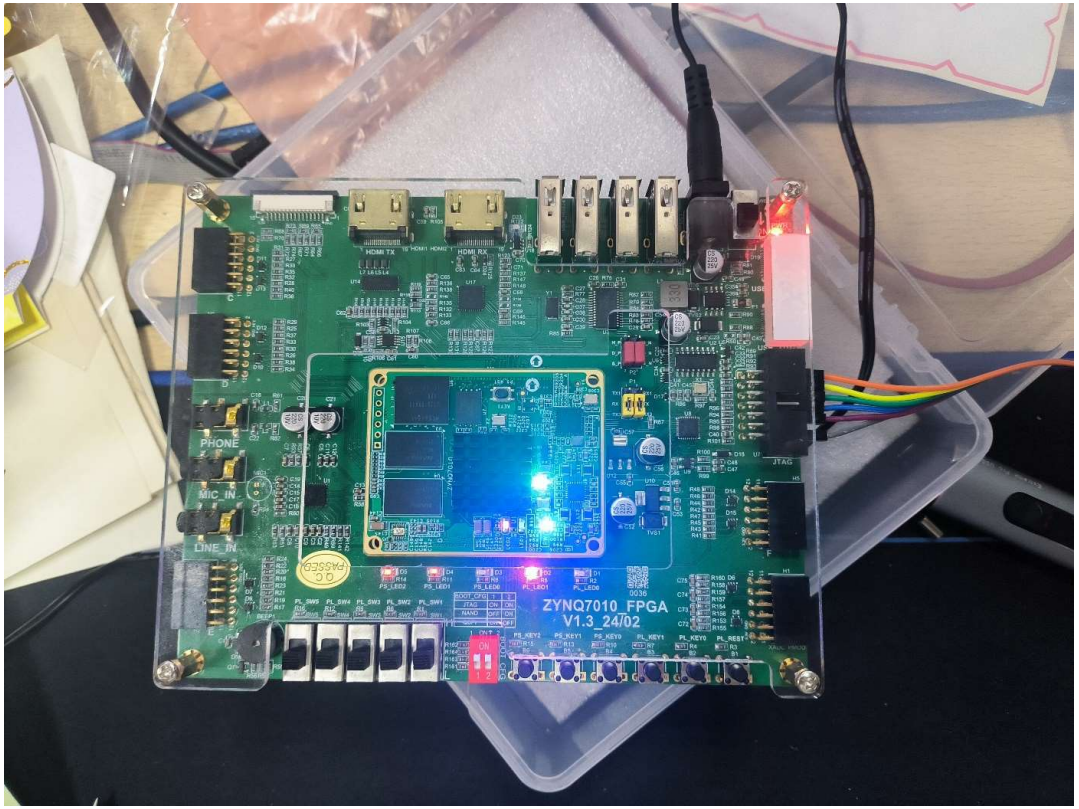
连接板卡与仿真器

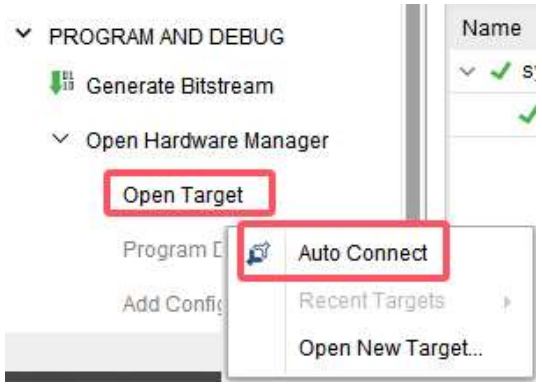


仿真器杜邦线的另一头带有防呆接口，直接接入板卡JTAG接口

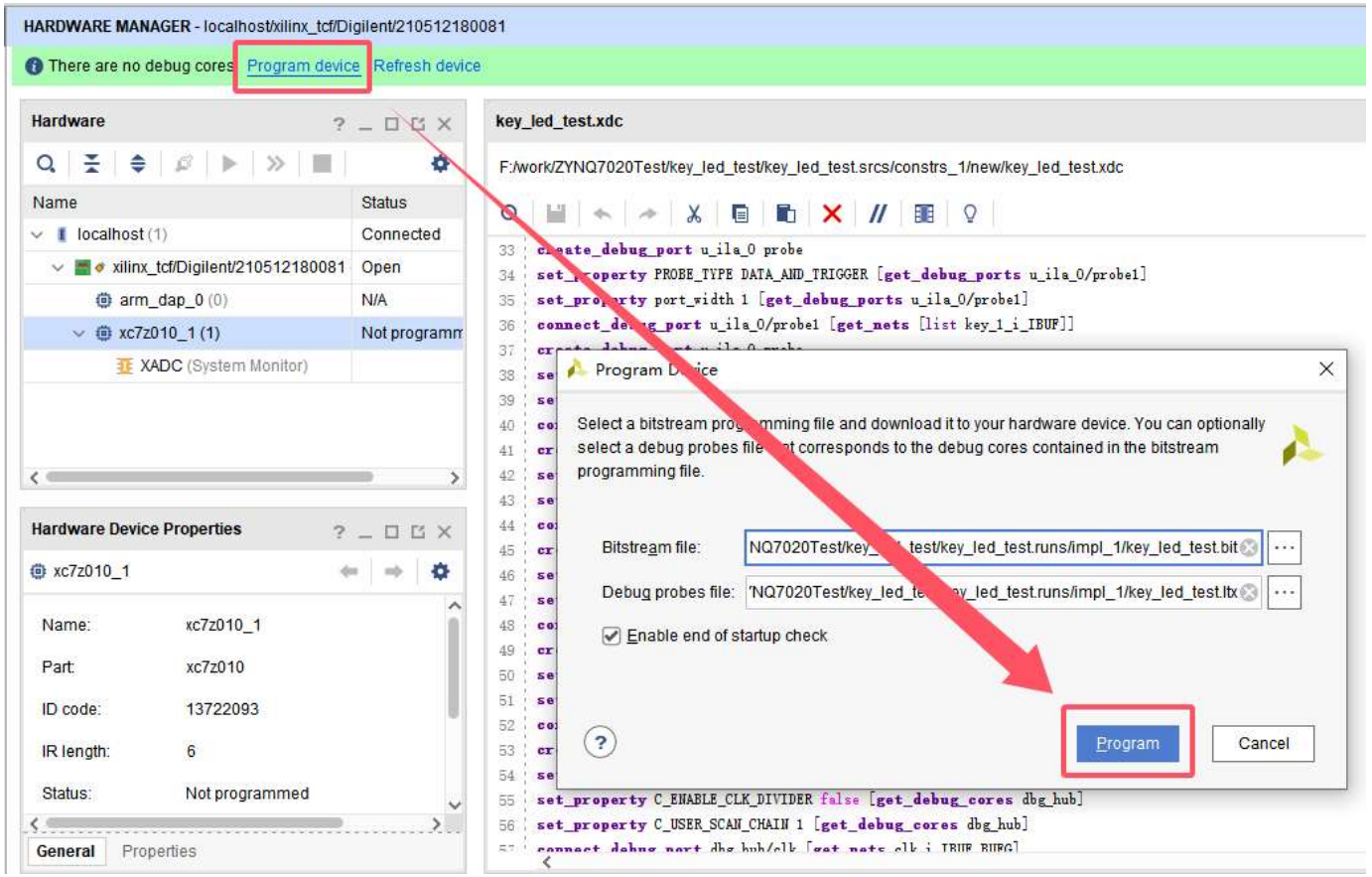
USB线接入USB UART接口提供供电，此时板卡电源开关不生效

比特流生成后，连接板卡准备烧录





烧录程序


HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210512180081
There are no debug cores
Program device
Refresh device
Hardware
Name
Status
localhost (1)
Connected
xilinx_tcf/Digilent/210512180081
Open
arm_dap_0 (0)
N/A
xc7z010_1 (1)
Not programm
XADC (System Monitor)
Hardware Device Properties
xc7z010_1
Name: xc7z010_1
Part: xc7z010
ID code: 13722093
IR length: 6
Status: Not programmed
General
Properties
key_led_test.xdc
F:/work/ZYNQ7020Test/key_led_test/key_led_test.srcs/constrs_1/new/key_led_test.xdc
Program Device
Select a bitstream programming file and download it to your hardware device. You can optionally select a debug probes file that corresponds to the debug cores contained in the bitstream programming file.
Bitstream file:
Debug probes file:
Enable end of startup check
Program
Cancel